MIL-S-19500/136A(NAVY)
4 February 1969
SUPERSEDING
MIL-S-19500/136(NAVY)
6 December 1960

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, GERMANIUM, LOW-POWER

TYPE 2N1310

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a low-power, NPN, germanium transistor and is in accordance with MIL-S-19500, except as otherwise specified herein.

1.2 Physical dimensions. See figure 1 (TO-9).

1.3 Maximum ratings.

P_T 1/ T_A = 25°C	V_{CBO}	V_{EBO}	V_{RT}	T_{stg}
mW	Vdc	Vdc	Vdc	°C
120	90	20	90	-65 to +85

1/Derate linearly 2 mW/°C for $T_A > 25°C$.

1.4 Primary electrical characteristics.

	h_{FE} V_{CE} = 0.25 Vdc I_C = 5 mAdc	h_{FE} V_{CE} = 0.25 Vdc I_C = 20 mAdc	I_{CBO} V_{CB} = 5 Vdc	f_{hfb} V_{CB} = 5 Vdc I_E = 1 mAdc
			μAdc	MHz
Min	20	30	---	1.0
Max	---	---	7	---

FSC 5961

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2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. Requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-S-19500.

3.3 Design, construction, and physical dimensions. *Transistors shall be of the design,* construction, and physical dimensions shown on figure 1.

3.4 Performance characteristics. Performance characteristics shall be as specified in tables I and II.

3.5 Marking. The following marking specified in MIL-S-19500 may be omitted from the body of the transistor at the option of the manufacturer:

(a) Country of origin.
(b) Manufacturer's designating symbol.

4. *QUALITY ASSURANCE PROVISIONS*

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall consist of the examinations and tests specified in tables I and II.

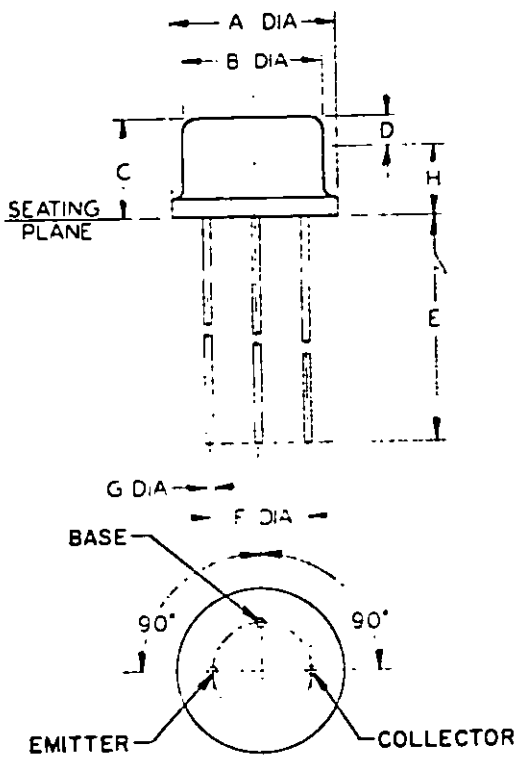
4.3 Quality conformance inspection. Quality conformance inspection shall consist of groups A and B inspections.

4.3.1 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table I.

4.3.2 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table II.

4.4 Methods of examination and test. Methods of examination and test shall be as specified in tables I and II.

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LTR	DIMENSIONS				NOTES
	INCHES		MILLIMETERS		
	MIN	MAX	MIN	MAX	
A	.290	.370	7.37	9.40	
B	.275	.335	6.99	8.51	
C	.200	.260	5.08	6.60	
D	.100	- - -	2.54	- - -	2
E	1.500	1.750	38.10	44.45	
F	.190	.210	4.83	5.33	
G	.016	.019	.41	.48	3,6
H	- - -	- - -	- - -	- - -	5

NOTES:
1. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
2. This zone is controlled for automatic handling. The variation in actual diameter within this zone shall not exceed .010 (.25 mm).
3. The specified lead diameter applies in the zone between .050 (1.27 mm) and .250 (6.35 mm) from the seating plane. Between .250 (6.35 mm) and 1.50 (38.10 mm) a maximum of .021 (.53 mm) diameter is held. Outside of these zones, the lead diameter is not controlled.
4. All leads shall be insulated from the case.
5. Details of outline in this zone optional.
6. 3 leads.

FIGURE 1. Physical dimensions of transistor type 2N1310 (TO-9).

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TABLE I. Group A inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			10				
Visual and mechanical examination	2071			---	---	---	---
Subgroup 2			5				
Breakdown voltage, collector to base	3001	Bias cond. D; I_C = 25 μAdc		BV_{CBO}	90	---	Vdc
Breakdown voltage, emitter to base	3026	Bias cond D; I_E = 25 μAdc		BV_{EBO}	20	---	Vdc
Collector to emitter voltage, reach through	---	V_{EB} = 1 Vdc		V_{RT}	90	---	Vdc
Collector to base cutoff current	3036	Bias cond. D; V_{CB} = 5 Vdc		I_{CBO}	---	7	μAdc
Emitter to base cutoff current	3061	Bias cond. D; V_{EB} = 5 Vdc		I_{EBO}	---	7	μAdc
Subgroup 3			5				
Forward-current transfer ratio	3076	I_C = 5 mAdc; V_{CE} = 0.25 Vdc		h_{FE}	20	---	---
Forward-current transfer ratio	3076	I_C = 20 mAdc; V_{CE} = 0.25 Vdc		h_{FE}	30	---	---
Base emitter voltage (nonsaturated)	3066	Test cond. B; I_C = 5 mAdc; V_{CE} = 0.25 Vdc		V_{BE}	---	0.5	Vdc
Base emitter voltage (nonsaturated)	3066	Test cond. B; I_C = 20 mAdc; V_{CE} = 0.25 Vdc		V_{BE}	---	0.6	Vdc
Subgroup 4			5				
Small-signal open-circuit reverse-voltage transfer ratio	3211	V_{CB} = 5 Vdc; I_E = 1 mAdc		h_{rb}	---	15	$x10^{-4}$
Small-signal open-circuit output admittance	3216	V_{CB} = 5 Vdc; I_E = 1 mAdc		h_{ob}	---	2.0	μmhos
Small-signal short-circuit forward-current transfer-ratio cutoff frequency	3301	V_{CB} = 5 Vdc; I_E = 1 mAdc		f_{hfb}	1.0	---	MHz
Subgroup 5			15				
High-temperature operation:		T_A = +71°C					
Collector to base cutoff current	3036	Bias cond. D; V_{CB} = 70 Vdc		I_{CBO}	---	300	μAdc
Low-temperature operation:		T_A = -55°C					
Forward-current transfer ratio	3076	I_C = 5 mAdc; V_{CE} = 0.25 Vdc		h_{FE}	12	---	---

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TABLE II. Group B inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			20				
Physical dimensions	2066	(See figure 1)		---	---	---	---
Subgroup 2			20				
Solderability	2026			---	---	---	---
Thermal shock (temperature cycling)	1051	Test cond. B except step 3 = +85°C		---	---	---	---
Thermal shock (glass strain)	1056	Test cond. A		---	---	---	---
Moisture resistance	1021			---	---	---	---
End points:							
Collector to base cutoff current	3036	Bias cond. D; V_{CB} = 5 Vdc		I_{CBO}	---	8.5	μAdc
Forward-current transfer ratio	3076	I_C = 5 mAdc; V_{CE} = 0.25 Vdc		h_{FE}	18	---	---
Subgroup 3			20				
Shock	2016	Nonoperating; 500 G; 1.0 msec, 5 blows in each orientation: X_1, Y_1, Y_2, and Z_1		---	---	---	---
Vibration fatigue	2046	Nonoperating		---	---	---	---
Vibration, variable frequency	2056	Nonoperating		---	---	---	---
Constant acceleration	2006	10,000 G; in each orientation: X_1, Y_1, Y_2, and Z_1		---	---	---	---
End points: (Same as subgroup 2)							
Subgroup 4			20				
Terminal strength (lead fatigue)	2036	Test cond. E		---	---	---	---
Subgroup 5			20				
Salt atmosphere (corrosion)	1041			---	---	---	---
End points: (Same as subgroup 2)							
Subgroup 6			λ = 10				
High-temperature life (nonoperating)	1031	T_{stg} = +85°C		---	---	---	---

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TABLE II. Group B inspection - Continued

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 6 - Continued							
End points:							
Collector to base cutoff current	3036	Bias cond. D; V_{CB} = 5 Vdc		I_{CBO}	---	10	µAdc
Forward-current transfer ratio	3076	I_C = 5 mAdc; V_{CE} = 0.25 Vdc		h_{FE}	16	---	---
Subgroup 7			λ = 10				
Steady-state operation life	1026			---	---	---	---
End points: (Same as subgroup 6)							

5. PREPARATION FOR DELIVERY

5.1 See MIL-S-19500, section 5.

6. NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

Preparing activity:
Navy - EC

(Project 5961-N178)

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